

# E-COMMERCE CUSTOMER ANALYTICS & PREDICTIVE INTELLIGENCE

## Level 3 – Advanced Project Report

**Organization:** Cognevance Technologies

**Project Title:** Big Data Analytics & Predictive Intelligence

**Domain:** Data Analytics, Big Data Analytics, and Machine Learning

**Dataset:** Olist Brazilian E-Commerce Public Dataset

**Tools & Technologies:** Python, Pandas, NumPy, PySpark, SQL, MySQL, Scikit-learn, Streamlit, Plotly, Power BI, Jupyter Notebook

**GitHub Repository:** <https://github.com/manavgawale4-tech/cognevance-bigdata-analytics>

---

## 1. Executive Summary

This project, titled **E-Commerce Customer Analytics & Predictive Intelligence**, focuses on analysing e-commerce transaction data to identify business trends, understand customer purchasing behaviour, evaluate delivery performance, and develop a machine learning model for delivery-time prediction.

The project uses the Olist Brazilian E-Commerce Public Dataset, which contains information about customers, orders, products, sellers, payments, reviews, and delivery timestamps.

Python and Pandas are used for data cleaning, transformation, and exploratory data analysis. SQL supports structured data analysis, while PySpark is used to perform distributed data processing and generate analytical summaries. A Random Forest Regression model is developed to estimate delivery time. An interactive dashboard built with Streamlit and Plotly presents important business KPIs and analytical visualisations.

The project demonstrates how data analytics and machine learning can support business decisions by identifying revenue trends, analysing customer retention, monitoring delivery performance, and highlighting areas that require improvement.

The model achieved a Mean Absolute Error (MAE) of 5.03 days, Root Mean Squared Error (RMSE) of 8.03 days, and  $R^2$  score of 0.2613 on the evaluated test data. These results indicate that the model has limited predictive performance and requires further improvement before being used for operational delivery estimates.

## 2. Problem Statement

E-commerce businesses generate large amounts of data through customer orders, product sales, payments, reviews, and deliveries. However, raw transaction data does not directly provide actionable business insights.

Businesses need to understand which product categories generate the most revenue, how sales change over time, whether customers place repeat orders, and how effectively orders are delivered.

They may also benefit from estimating delivery times in advance to improve planning and customer communication.

The main problem addressed by this project is the transformation of raw e-commerce data into meaningful business insights using data preprocessing, SQL, distributed data processing, interactive visualisation, and machine learning.

### **3. Project Objectives**

The main objectives of this project are:

1. To collect and analyse a large, multi-table e-commerce dataset.
2. To clean and preprocess data containing missing values, duplicate records, and inconsistent data types.
3. To perform exploratory data analysis using Python, Pandas, and NumPy.
4. To use SQL queries for structured business analysis.
5. To apply PySpark for large-scale data processing and analytical summaries.
6. To calculate business KPIs related to orders, customers, revenue, and delivery performance.
7. To analyse monthly and yearly sales trends.
8. To understand customer purchasing and repeat-order behaviour.
9. To develop a Random Forest Regression model for delivery-time prediction.
10. To create an interactive dashboard for exploring important business metrics.
11. To provide business recommendations based on the analytical results.
12. To document the project's architecture, workflow, limitations, and future improvements.

### **4. Dataset Description**

#### **4.1 Dataset Overview**

The project uses the Olist Brazilian E-Commerce Public Dataset. The dataset contains multiple CSV files representing different aspects of e-commerce operations.

These files are connected through identifiers such as order ID, customer ID, product ID, and seller ID. Joining these tables makes it possible to analyse relationships between orders, customers, products, payments, and deliveries.

#### 4.2 Main Dataset Files

| Dataset File                          | Description                                                            |
|---------------------------------------|------------------------------------------------------------------------|
| olist_orders_dataset.csv              | Order status and purchase, approval, shipping, and delivery timestamps |
| olist_customers_dataset.csv           | Customer identifiers and location information                          |
| olist_order_items_dataset.csv         | Products included in orders, seller IDs, prices, and freight values    |
| olist_order_payments_dataset.csv      | Payment methods, payment values, and instalment information            |
| olist_order_reviews_dataset.csv       | Customer reviews and review scores                                     |
| olist_products_dataset.csv            | Product identifiers, categories, and product attributes                |
| olist_sellers_dataset.csv             | Seller identifiers and location information                            |
| olist_geolocation_dataset.csv         | Geographic coordinates associated with postal-code prefixes            |
| product_category_name_translation.csv | Translation of product category names into English                     |

#### 4.3 Dataset Size

The project report records the following dataset sizes:

| Dataset     | Number of Records |
|-------------|-------------------|
| Orders      | 99,441            |
| Order Items | 112,650           |
| Products    | 32,951            |

| Dataset | Number of Records |
|---------|-------------------|
|---------|-------------------|

|           |        |
|-----------|--------|
| Customers | 99,441 |
|-----------|--------|

|                       |    |
|-----------------------|----|
| Category Translations | 71 |
|-----------------------|----|

The files have different numbers of records because each file represents a different entity or relationship. For example, one order may contain multiple products, and therefore the number of order-item records can exceed the number of orders.

## 5. Technologies Used

| Technology              | Purpose                                                 |
|-------------------------|---------------------------------------------------------|
| Python                  | Main programming language for analytics and modelling   |
| Pandas                  | Data loading, cleaning, transformation, and aggregation |
| NumPy                   | Numerical operations and model evaluation calculations  |
| SQL                     | Structured querying and business analysis               |
| MySQL                   | Relational database storage and query execution         |
| PySpark                 | Distributed data processing and large-scale analytics   |
| Scikit-learn            | Machine learning model development and evaluation       |
| Random Forest Regressor | Delivery-time prediction                                |
| Joblib                  | Saving and loading the trained machine learning model   |
| Streamlit               | Development of the interactive analytics dashboard      |
| Plotly                  | Interactive charts and visualisations                   |
| Jupyter Notebook        | Exploratory analysis and model development              |
| Visual Studio Code      | Project development and code management                 |
| Power BI                | Dashboard file included in the project repository       |

## 6. Data Cleaning and Preprocessing

Data preprocessing is an important stage because inaccurate, incomplete, or inconsistent data can affect analytical results and machine learning performance.

The following preprocessing activities were used in the project.

### 6.1 Data Loading

The dataset CSV files were loaded into Python using Pandas. Initial inspection was performed to understand the structure, column names, data types, and record counts.

## **6.2 Missing-Value Handling**

Missing values were examined in important columns. Missing or invalid values were handled according to the requirements of each analysis.

For example, delivery-time calculations require valid purchase and delivery timestamps. Records without the required timestamps cannot be used for that particular calculation.

## **6.3 Duplicate-Record Checking**

Duplicate records were checked to reduce the risk of counting the same record more than once. Duplicate removal was applied where appropriate to the relevant data.

## **6.4 Date-Time Conversion**

Order-related timestamps were converted into appropriate date-time formats. This enabled calculations such as delivery duration and grouping sales by month or year.

## **6.5 Data Integration**

Related tables were joined using appropriate identifiers. These joins allowed customer, order, product, payment, and delivery information to be analysed together.

## **6.6 Feature Engineering**

Additional analytical features were derived from the original columns, including:

- Purchase month
- Purchase year
- Monthly sales period
- Delivery duration in days
- Customer order frequency
- Delivery performance category

For machine learning, the selected features included product ID, seller ID, product price, freight value, and purchase month.

## **6.7 Data Aggregation**

Data was grouped by relevant business dimensions such as month, year, product category, customer, and delivery status. Aggregations were used to calculate revenue, order counts, customer behaviour, and delivery-performance summaries.

## 7. SQL Analytics

SQL was used to perform structured queries on e-commerce data and extract business information.

The SQL analysis included queries for:

1. Total order counts.
2. Unique customer counts.
3. Delivered-order counts.
4. Revenue calculations.
5. Monthly sales trends.
6. Yearly sales trends.
7. Revenue by product category.
8. Top-performing product categories.
9. Customer order frequency.
10. Repeat-customer analysis.
11. Average delivery time.
12. On-time and late delivery analysis.
13. Business KPI summaries.

These queries help convert raw relational data into structured outputs that can be used for reporting and business decision-making.

## 8. Key Business KPIs

The following KPI values are recorded in the project analysis.

| KPI                       | Recorded Value    |
|---------------------------|-------------------|
| Total Orders              | 99,441            |
| Unique Customers          | 96,096            |
| Delivered Orders          | 96,478            |
| Delivered Product Revenue | BRL 13,221,498.11 |
| Average Delivery Time     | 12.56 days        |

| KPI | Recorded Value |
|-----|----------------|
|-----|----------------|

|                                 |        |
|---------------------------------|--------|
| Customers with Delivered Orders | 93,358 |
|---------------------------------|--------|

|                  |       |
|------------------|-------|
| Repeat Customers | 2,801 |
|------------------|-------|

|                      |       |
|----------------------|-------|
| Repeat-Customer Rate | 3.00% |
|----------------------|-------|

**Note:** KPI values depend on the filtering rules, date conditions, and definitions used in the analysis. Customer counts, order counts, and revenue totals may differ across sections when they use different subsets of records.

## 9. Monthly and Yearly Sales Analysis

Sales trend analysis helps identify changes in business activity over time.

### 9.1 Monthly Sales Analysis

Monthly sales were calculated by grouping the relevant order and product data by month.

The recorded analysis identifies **November 2017** as the peak month, with revenue of **BRL 987,765.37** and **7,289 delivered orders**.

Monthly analysis can help a business:

- Identify periods of high sales activity.
- Compare revenue across months.
- Recognise seasonal patterns.
- Plan inventory and staffing requirements.
- Evaluate changes in order volume.

### 9.2 Yearly Sales Analysis

The report records the following yearly figures.

| Year | Recorded Revenue (BRL) | Delivered Orders |
|------|------------------------|------------------|
|------|------------------------|------------------|

|      |           |     |
|------|-----------|-----|
| 2016 | 40,470.98 | 267 |
|------|-----------|-----|

|      |              |        |
|------|--------------|--------|
| 2017 | 5,962,902.01 | 43,428 |
|------|--------------|--------|

|      |              |        |
|------|--------------|--------|
| 2018 | 7,218,125.12 | 52,783 |
|------|--------------|--------|

These values show substantial growth in the recorded revenue and delivered-order totals between the periods shown. However, the 2016 and 2018 totals may represent

partial periods within the dataset. Therefore, the figures should not automatically be interpreted as complete calendar-year comparisons.

**10. Product Category Revenue Analysis**

Product category analysis helps identify which categories contribute the most to sales.

The following categories were identified among the leading revenue categories in the recorded analysis.

| Product Category      | Revenue (BRL) | Items  |
|-----------------------|---------------|--------|
| Health & Beauty       | 1,233,131.72  | 9,465  |
| Watches and Gifts     | 1,166,176.98  | 5,859  |
| Bed Bath Table        | 1,023,434.76  | 10,953 |
| Sports Leisure        | 954,852.55    | 8,431  |
| Computers Accessories | 888,724.61    | 7,644  |

The analysis indicates that revenue contribution differs across product categories. Businesses can use this information to plan category-level promotions, inventory allocation, and marketing activities.

Revenue alone does not necessarily indicate profitability. Additional analysis of costs, discounts, returns, and profit margins would be required before making investment decisions about individual categories.

**11. Customer Behaviour Analysis**

Customer behaviour analysis focuses on order frequency and repeat purchasing.

The recorded analysis contains the following figures:

| Customer Metric                 | Recorded Value |
|---------------------------------|----------------|
| Customers with Delivered Orders | 93,358         |
| Repeat Customers                | 2,801          |
| Single-Order Customers          | 90,557         |
| Repeat-Customer Rate            | 3.00%          |

Repeat-customer analysis helps businesses understand whether customers return to make additional purchases.

The recorded repeat-customer rate suggests that repeat purchasing is an important area for further investigation. However, the result should be interpreted in the context of the dataset's observation period and the definition used for a repeat customer.

Potential business actions include improving post-purchase communication, offering relevant recommendations, and examining customer feedback to identify opportunities to improve customer retention.

**12. Delivery Performance Analysis**

Delivery performance is an important part of the e-commerce customer experience. Delays may affect customer satisfaction, reviews, and repeat purchasing.

**12.1 Average Delivery Time**

The recorded average delivery time was **12.56 days**, calculated from **96,470 delivered orders** with valid purchase and delivery timestamps.

Delivery duration can be calculated by finding the difference between the purchase timestamp and the delivery timestamp.

**12.2 On-Time and Late Deliveries**

The recorded delivery-performance results are:

| <b>Delivery Status</b> | <b>Orders</b> |
|------------------------|---------------|
| On Time or Early       | 88,644        |
| Late                   | 7,826         |

Total with Delivery Classification 96,470

The analysis distinguishes orders delivered on time or early from those delivered late.

This information can help businesses investigate delivery delays, identify operational issues, and improve customer communication.

**13. Machine Learning Model: Delivery-Time Prediction**

**13.1 Objective**

The objective of the machine learning component is to estimate delivery time in days using selected product, seller, pricing, freight, and purchase-month features.

The problem is treated as a regression task because the target variable, delivery duration, is numeric.

**13.2 Selected Features**

The model uses the following input features:

| Feature        | Description                            |
|----------------|----------------------------------------|
| product_id     | Encoded product identifier             |
| seller_id      | Encoded seller identifier              |
| price          | Product price                          |
| freight_value  | Freight charge                         |
| purchase_month | Month in which the order was purchased |

The target variable is **delivery\_time\_days**.

Product and seller identifiers were converted into numeric category codes to make them usable by the model.

### 13.3 Model Selection

A Random Forest Regressor was selected for delivery-time prediction.

Random Forest is an ensemble learning algorithm that combines the predictions of multiple decision trees. For regression, the model combines the outputs of the trees to produce a numeric estimate.

The model configuration used in the project was:

| Parameter          | Value                   |
|--------------------|-------------------------|
| Algorithm          | Random Forest Regressor |
| Number of Trees    | 100                     |
| Maximum Tree Depth | 15                      |
| Random State       | 42                      |
| Number of Jobs     | -1                      |
| Test Set Size      | 20%                     |

### 13.4 Training Procedure

The selected features were separated from the target variable. The data was divided into training and testing sets using an 80:20 split and a fixed random state.

The model was trained on the training data and evaluated on the testing data.

The trained model was saved using Joblib in the following file:

models/delivery\_time\_model.pkl

The saved model was subsequently loaded and tested with sample feature values to verify that it could generate a prediction.

### 13.5 Model Evaluation

The model was evaluated using three common regression metrics.

| Evaluation Metric              | Recorded Result |
|--------------------------------|-----------------|
| Mean Absolute Error (MAE)      | 5.03 days       |
| Root Mean Squared Error (RMSE) | 8.03 days       |
| R <sup>2</sup> Score           | 0.2613          |

#### Mean Absolute Error (MAE)

MAE measures the average absolute difference between the actual delivery times and the predicted delivery times. The recorded MAE of 5.03 days indicates that the predictions differed from actual values by an average absolute amount of approximately 5.03 days on the evaluated test set.

#### Root Mean Squared Error (RMSE)

RMSE measures prediction errors while assigning greater weight to larger errors. The recorded RMSE was 8.03 days.

#### R<sup>2</sup> Score

The R<sup>2</sup> score indicates how much of the variation in the target variable is explained by the model relative to a baseline prediction.

The recorded R<sup>2</sup> score of 0.2613 indicates limited explanatory performance. Although the model produces delivery-time estimates, its current predictive performance is not strong enough to support claims of highly accurate delivery prediction.

### 13.6 Model Limitations

The current model has several limitations:

1. The selected features may not capture important causes of delivery delays.
2. Product and seller identifiers are encoded as category codes, which do not represent meaningful numerical distances.
3. A random train-test split may not fully reflect how the model would perform on future orders.

4. The model's  $R^2$  score indicates that much of the variation in delivery time remains unexplained.
5. Further feature engineering, validation, and model comparison are required before operational use.

Potential improvements include incorporating order approval time, shipping date, seller location, customer location, distance-related features, order characteristics, and other relevant variables where available.

## **14. PySpark Analytics**

PySpark was used to demonstrate distributed data processing for e-commerce analytics.

The PySpark workflow loads relevant datasets, performs preprocessing, joins related tables, derives analytical fields, and generates grouped summaries.

The analysis includes:

- Monthly sales summaries.
- Yearly sales summaries.
- Revenue by product category.
- Customer order-frequency analysis.
- Delivery-duration summaries.
- Delivery-performance classification.

The workflow generates analytical CSV outputs, including:

- monthly\_sales.csv
- category\_revenue.csv
- customer\_behavior.csv
- delivery\_summary.csv
- yearly\_sales.csv
- delivery\_performance.csv

These outputs can be used for reporting, visualisation, and further business analysis.

## **15. Interactive Analytics Dashboard**

### **15.1 Dashboard Overview**

An interactive dashboard was developed using Streamlit, Pandas, and Plotly to present key e-commerce performance indicators.

The dashboard allows users to explore selected data through filters and interactive visualisations.

### **15.2 Dashboard Features**

The dashboard includes the following analytical components:

1. Total orders.
2. Delivered orders.
3. Unique customers.
4. Revenue for the selected category or filtered dataset.
5. Monthly revenue trends.
6. Top product categories.
7. Repeat and single-order customer analysis.
8. Payment analysis.
9. Delivery-performance analysis.
10. Average delivery-time analysis and on-time delivery rate.

### **15.3 Interactive Filters**

The dashboard includes filters for exploring different portions of the dataset, such as dates and product categories, depending on the available controls.

When a user changes a filter, the related KPIs and visualisations update to reflect the selected data.

### **15.4 Dashboard Technology**

Streamlit provides the application interface, Pandas handles the tabular data, and Plotly creates interactive charts.

The project repository also contains a Power BI dashboard file named `Ecommerce_Sales_Dashboard.pbix`.

## **16. Business Insights and Recommendations**

The analysis provides several potential business insights.

### **16.1 Revenue Trends**

Monthly and yearly revenue summaries help identify periods of increased sales activity.

**Recommendation:** Use historical sales trends to support inventory planning, staffing decisions, and the scheduling of promotional campaigns. Confirm seasonal patterns across multiple complete periods before treating them as recurring trends.

## **16.2 Product Category Performance**

The recorded category analysis identifies Health & Beauty, Watches and Gifts, and Bed Bath Table among the leading categories by revenue.

**Recommendation:** Review category-level revenue alongside profit margins, discounts, product availability, and customer demand before allocating additional marketing resources.

## **16.3 Customer Retention**

The recorded repeat-customer rate is 3.00% under the project's selected customer definition and dataset period.

**Recommendation:** Investigate repeat purchasing by customer cohort, acquisition period, category, and time since first purchase. Consider relevant product recommendations and post-purchase communication to encourage retention.

## **16.4 Delivery Performance**

The recorded analysis identifies 7,826 late deliveries among 96,470 orders with delivery classifications.

**Recommendation:** Examine late deliveries by seller, destination, order period, and shipping characteristics. Use the findings to identify possible operational bottlenecks and improve delivery communication.

## **16.5 Predictive Intelligence**

The Random Forest model generates delivery-time estimates, but its current evaluation results indicate limited predictive performance.

**Recommendation:** Improve the feature set, compare alternative regression models, and validate predictions using a time-based holdout before considering the model for operational decision-making.

## **16.6 Data-Driven Decision-Making**

The combination of SQL analysis, Python processing, PySpark summaries, machine learning, and interactive dashboards provides different ways to explore the same business problem.

**Recommendation:** Establish consistent KPI definitions and data-refresh procedures so that business users can interpret metrics reliably across reports and dashboards.

## **17. System Architecture and Workflow**

The project follows a data analytics workflow that transforms raw data into business insights and predictive outputs.

### **17.1 Workflow Stages**

#### **Stage 1: Data Collection**

Collect the e-commerce CSV files containing orders, customers, products, sellers, payments, and reviews.

#### **Stage 2: Data Loading**

Load the required datasets into Python and, where applicable, relational database or distributed-processing environments.

#### **Stage 3: Data Preprocessing**

Inspect data types, handle missing or invalid values, check duplicates, convert timestamps, and prepare the relevant columns.

#### **Stage 4: Data Integration**

Join related datasets using suitable identifiers to create analysis-ready data.

#### **Stage 5: Exploratory Data Analysis**

Use Pandas, NumPy, and visualisation tools to explore sales trends, product categories, customer behaviour, and delivery performance.

#### **Stage 6: SQL and PySpark Analytics**

Run structured queries and distributed data-processing workflows to calculate business summaries.

#### **Stage 7: Machine Learning**

Prepare the model features and target variable, split the data, train the Random Forest Regressor, evaluate its performance, and save the trained model.

#### **Stage 8: Dashboard Development**

Present business KPIs and analytical charts through interactive dashboard components.

#### **Stage 9: Reporting and Recommendations**

Summarise the findings, discuss limitations, and provide business recommendations.

### **17.2 Architecture Summary**

Raw E-Commerce CSV Files

↓

Data Loading and Preprocessing

↓

Data Integration and Feature Engineering

↓

SQL Analytics, Python Analysis, and PySpark Processing

↓

Business KPIs and Analytical Outputs

↓

Machine Learning Model and Interactive Dashboard

↓

Business Insights, Recommendations, and Final Report

## **18. Limitations and Future Improvements**

### **18.1 Current Limitations**

1. The model's  $R^2$  score indicates limited predictive performance.
2. The selected model features may not fully represent the factors affecting delivery time.
3. Historical dataset patterns may not represent current e-commerce operations.
4. Revenue and customer metrics depend on the filters and definitions used.
5. The analysis does not by itself establish causal relationships between customer behaviour, delivery performance, and revenue.
6. The current project is an analytical prototype and does not demonstrate a complete production deployment or real-time prediction service.

### **18.2 Future Improvements**

Possible improvements include:

- Adding more relevant features to the delivery-time model.
- Comparing Random Forest with other regression algorithms.
- Using time-based validation to evaluate predictions on later orders.
- Improving customer segmentation and cohort analysis.
- Analysing reviews alongside delivery performance.
- Adding profit-margin and discount analysis.
- Automating data refresh and report generation.
- Improving dashboard navigation and export options.
- Establishing consistent KPI definitions across SQL, Python, PySpark, and dashboards.

- Deploying the dashboard in an appropriate hosted environment after additional testing.

## 19. Conclusion

The **E-Commerce Customer Analytics & Predictive Intelligence** project demonstrates the application of data analytics, SQL, distributed data processing, machine learning, and interactive visualisation to an e-commerce dataset.

The project analyses revenue trends, product category performance, customer purchasing behaviour, and delivery operations. It also develops a Random Forest Regression model to estimate delivery time and presents key analytical findings through an interactive dashboard.

The recorded model evaluation results—MAE of 5.03 days, RMSE of 8.03 days, and  $R^2$  of 0.2613—show that the current model provides a starting point for predictive analysis but requires further improvement.

Overall, the project demonstrates an end-to-end analytical workflow, from raw data processing to business reporting and predictive modelling. It also highlights the importance of data quality, reliable KPI definitions, honest model evaluation, and continuous improvement when using analytics to support business decisions.

## 20. Project Repository

The project source code, datasets, notebooks, analytical scripts, model file, dashboard files, and documentation are organised in the project repository.

### GitHub Repository:

<https://github.com/manavgawale4-tech/cogneance-bigdata-analytics>

The repository provides access to the project artefacts and supports further review and development of the analytical workflow.

---

**End of Report**